



DEEPSEEK-V2

DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

Efficient MoE Language Model with Multi-Head Latent Attention

236B total / 21B activated

128K context length

ArXiv: 2405.04434

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference

Needed: An architecture that decouples model size from activated parameters to achieve strong intelligence economically.



Prohibitive Training Costs

Larger models yield better intelligence, but scaling dense architectures incurs exponential increases in compute costs.

100B+ Parameters

Dense model training requires massive cluster hours

Constraint: Budget limits scaling frequency



Inference Speed Bottlenecks

During autoregressive generation, conventional attention requires caching full Key-Value pairs per token per head.

KV Cache Memory Wall

Limits batch size and maximum generation throughput

Constraint: Deployment is economically unviable

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block

Novel Transformer block design enables training 236B parameters while activating only 21B per token.

Architectural Innovations

1. Multi-Head Latent Attention (MLA)

- Replaces standard Multi-Head Attention
- Compresses KV cache into a latent vector
- Decoupled RoPE for positional encoding

2. DeepSeekMoE

- Replaces dense Feed-Forward Network
- Isolates shared experts for common knowledge
- Sparsely activates routed fine-grained experts

3. Extreme Sparsity

- 236B total parameters
- Only 21B activated during inference
- Decouples capability from compute cost

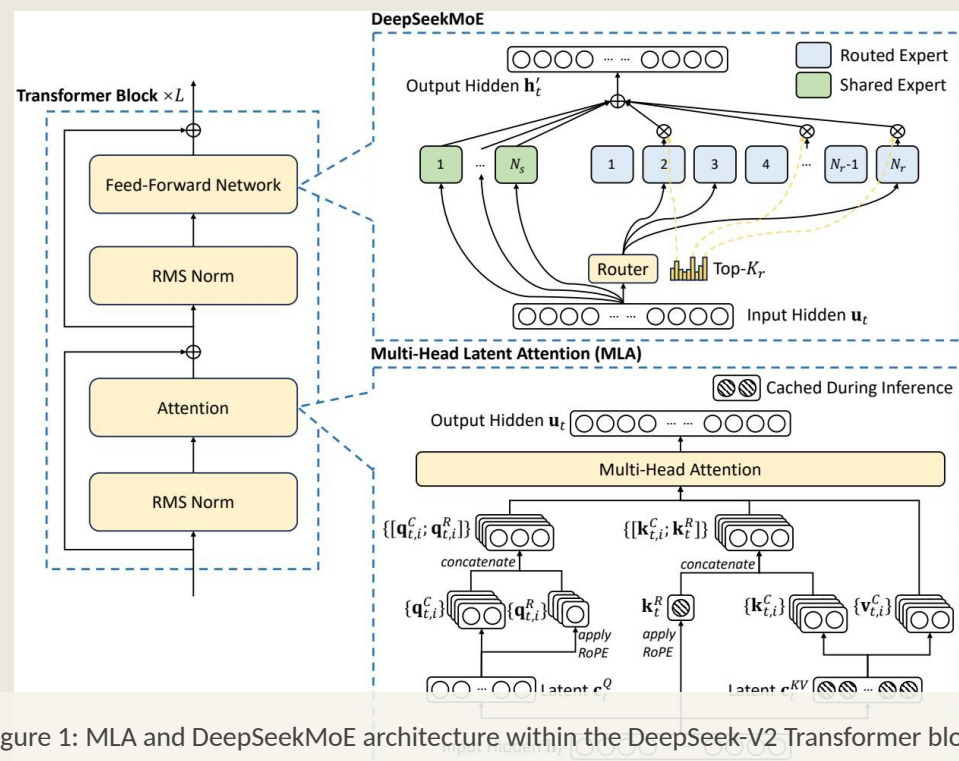
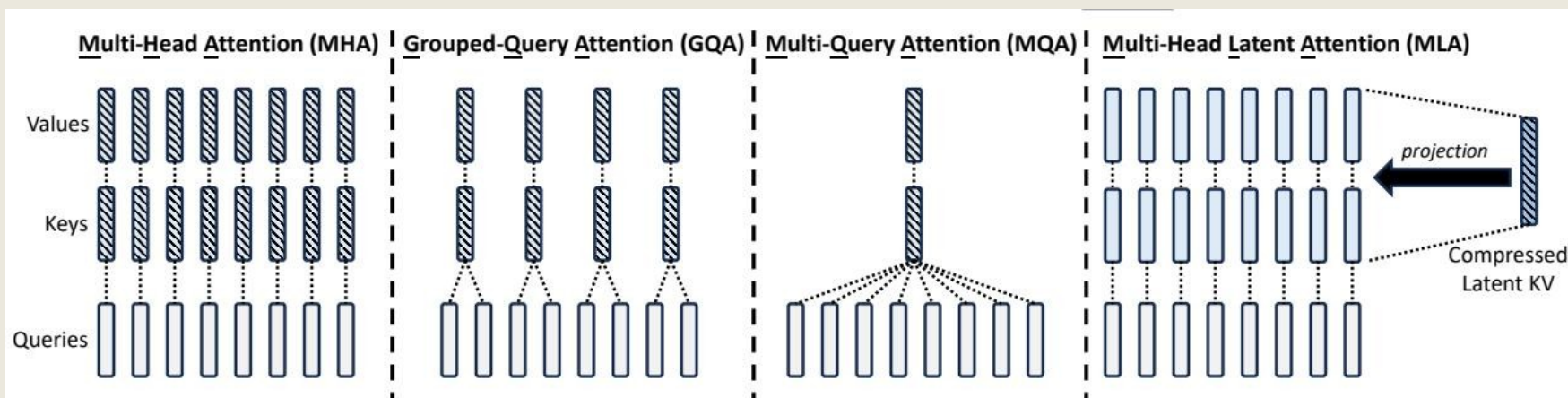


Figure 1: MLA and DeepSeekMoE architecture within the DeepSeek-V2 Transformer block.

Multi-Head Latent Attention Compresses KV Cache by 93.3%

MLA solves the KV cache bottleneck via low-rank key-value joint compression while maintaining MHA performance.



Low-Rank Compression

Instead of caching full K and V vectors, MLA compresses them into a single, compact latent vector.

On-the-Fly Reconstruction

During inference, full keys and values are quickly reconstructed from the cache via learned matrices.

Superior Efficiency

Drastically smaller cache footprint than GQA/MQA, while outperforming standard Multi-Head Attention.

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Separating shared experts and routing fine-grained experts optimizes load balancing and cross-node communication.

Shared Experts

Always activated to capture common knowledge across all domains.

Routed Experts

Sparsely activated via a top-K router to handle specialized domain knowledge.

Fine-Grained Segmentation

Smaller expert sizes increase specialization potential without adding parameters.

Device-Limited Routing

Constrains tokens to experts on at most M devices to reduce communication overhead.

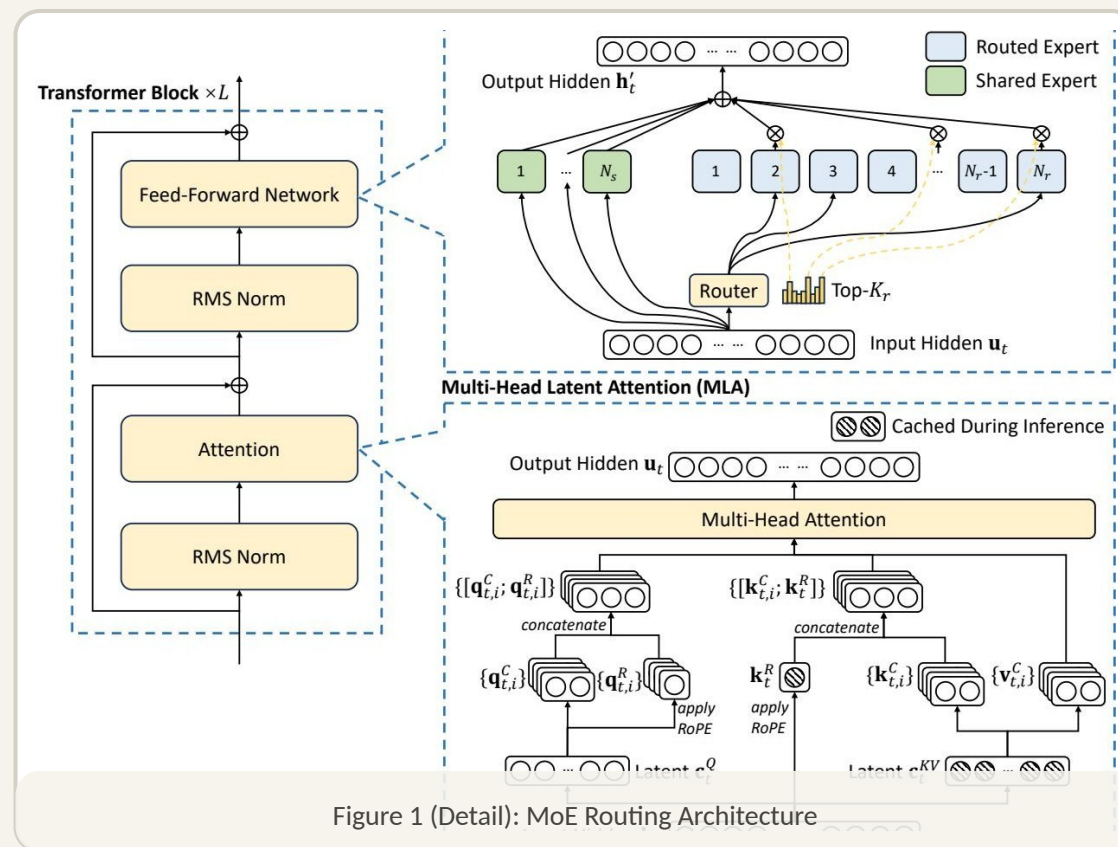


Figure 1 (Detail): MoE Routing Architecture

EFFICIENCY GAINS

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. De

DeepSeek-V2 shatters the trade-off between model capacity and operational efficiency.

42.5%

Training Cost Reduction

DeepSeekMoE dramatically lowers compute requirements while training 236B total parameters.

93.3%

KV Cache Reduction

Multi-Head Latent Attention (MLA) compresses key-value vectors, freeing up GPU memory.



5.76×

Higher Throughput

Reduced memory wall allows for vastly larger batch sizes during autoregressive generation.

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated

DeepSeek-V2 occupies a Pareto-optimal position, matching or exceeding 70B-class dense models.

Pareto-Optimal Efficiency

DeepSeek-V2 Score:

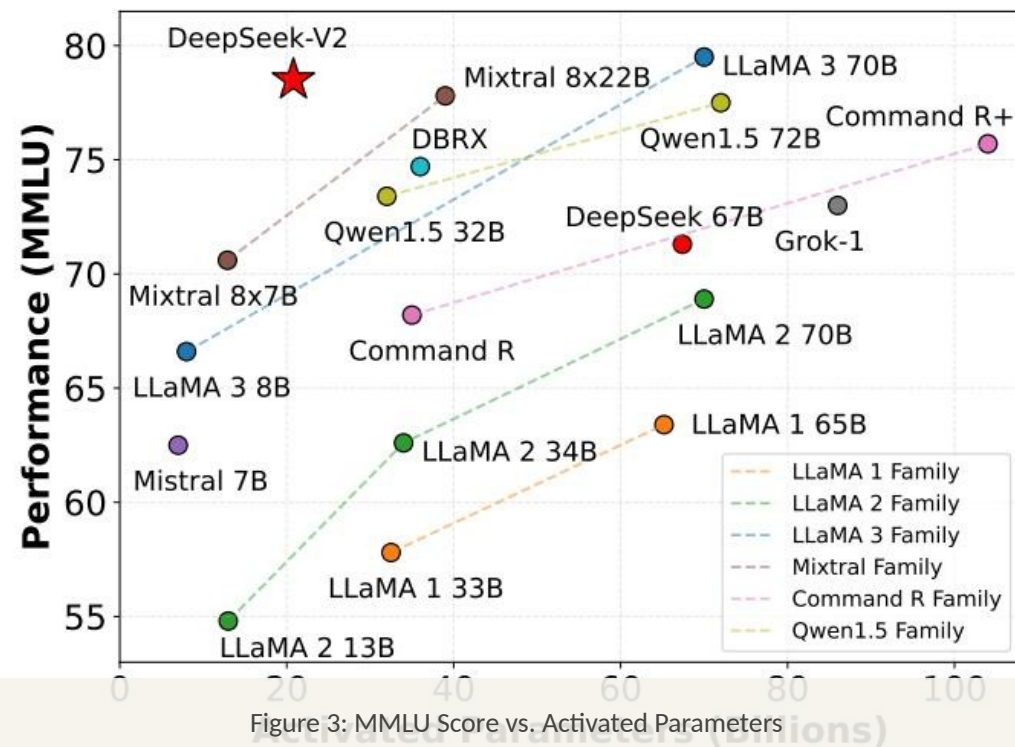
78.5 on MMLU

Unmatched Efficiency:

Achieved with only 21B activated parameters.

Baseline Comparisons:

- LLaMA 3 70B (77.0)
- Mixtral 8x22B (77.6, 39B act)
- Qwen1.5 72B (75.6)



DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Bench

Through SFT and GRPO-based RL, DeepSeek-V2 Chat outperforms all open-source models on AlignBench.



Alignment Pipeline

Supervised Fine-Tuning (SFT) on 1.5M conversational sessions, followed by Reinforcement Learning via Group Relative Policy Optimization (GRPO)

Benchmark	DeepSeek-V2 Chat (RL) Score	Performance Context
AlpacaEval 2.0	38.9%	Length-controlled win rate in English open-ended evaluation.
MT-Bench (Overall)	8.97	Competitive with closed-source systems.
AlignBench (Chinese)	7.91	Outperforms all open-source models.

CONCLUSION

Key Takeaways: Architectural Innovation Unlocks Efficient

MLA Innovation

Multi-Head Latent Attention is a superior KV-cache solution, **reducing footprint by 93.3%** over MHA.

DeepSeekMoE

Economical training via fine-grained experts and shared expert isolation, **cutting training costs by 42.5%**.

Pareto Efficiency

Top-tier MMLU performance achieved at a fraction of the compute, **activating only 21B parameters**.

Strong Alignment

DeepSeek-V2 Chat demonstrates exceptional multilingual and open-ended conversation capabilities, **dominating Chinese**.